

OPEN INTEREST AND OPTION GREEKS

Open Interest (OI)

- Open interest refers to the total number of outstanding (unliquidated) futures or options contracts in the market at a given point in time. It represents the total number of contracts that have not been closed or fulfilled through an offsetting transaction.
- Open interest increases when new contracts are created through new trades.
- Open interest decreases when contracts are closed out or expire.

Open Interest (OI)

- OI provides information about the liquidity and activity in a particular futures or options contract.
- Higher open interest generally indicates greater market participation and better liquidity.

How to Interpret OI

- In practice, OI is often analysed more from the seller's (writer's) point of view than the buyer's.
- CE sellers generally expect the market to go down or not rise much.
- PE sellers generally expect the market to go up or not fall much.

OI, Support & Resistance

- Look for the maximum OI in OTM strikes.
- High CE OI above the CMP can act as a resistance.
- High PE OI below the CMP can act as a support

Option Chain (Equity Derivatives)

[Futures contracts](#)

View Options Contracts for:

Select Symbol

Expiry Date

Strike Price

NIFTY

OR

Select

20-Jan-2026

OR

Select

Underlying Index : **NIFTY 25,694.35** As on 16-Jan-2026 15:30:00 IST

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	CALLS											PUTS										
	OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI	
	54	10	94	-	859.25	14.40	65	868.25	873.90	260	24,850.00	325	2.50	2.60	2,210	-4.60	2.45	15.49	3,93,568	-774	10,155	
	147	70	197	15.79	825.50	37.30	65	817.15	822.80	65	24,900.00	2,015	2.70	2.85	650	-5.10	2.70	14.93	7,17,516	16,267	41,587	
	216	5	34	-	761.95	18.40	260	768.75	774.20	260	24,950.00	650	3.10	3.20	975	-5.40	3.20	14.51	5,29,434	3,475	20,773	
	2,116	-466	4,647	-	720.70	28.55	65	720.60	723.20	260	25,000.00	845	3.70	3.85	1,560	-6.05	3.70	14.03	15,35,440	-5,602	1,12,938	
	402	166	606	10.55	672.55	27.90	65	670.95	674.15	260	25,050.00	65	4.20	4.35	1,365	-6.75	4.20	13.48	5,77,856	7,417	23,339	
	647	-249	1,255	9.84	622.60	24.35	65	621.65	623.75	325	25,100.00	390	4.70	4.80	585	-7.80	4.70	12.89	11,16,966	2,258	54,603	
	407	22	1,247	10.59	574.10	25.30	65	572.45	574.00	65	25,150.00	715	5.35	5.45	260	-8.80	5.35	12.31	7,13,136	3,875	23,442	
	1,283	-106	8,996	9.57	523.85	25.65	65	523.40	525.15	260	25,200.00	65	6.10	6.20	910	-10.40	6.10	11.72	12,92,482	1,374	62,169	
	538	-16	2,111	10.60	477.55	23.70	390	474.45	476.80	325	25,250.00	975	7.20	7.25	975	-12.45	7.20	11.18	9,39,999	28,590	52,311	
	1,689	-400	12,622	9.84	428.35	20.25	260	426.85	428.30	390	25,300.00	780	9.10	9.20	1,040	-14.60	9.15	10.80	18,79,073	31,781	88,695	
	746	-140	5,660	9.46	380.00	16.05	260	379.85	381.45	260	25,350.00	12,675	11.90	11.95	910	-17.45	11.90	10.47	13,17,951	8,666	36,382	
	2,875	-121	52,627	9.35	333.80	13.85	390	333.75	335.00	1,430	25,400.00	975	15.90	16.05	1,170	-20.50	15.90	10.20	27,62,720	26,177	92,081	
	1,369	-41	32,399	9.54	290.95	11.35	260	289.50	290.70	325	25,450.00	975	21.15	21.25	195	-24.00	21.15	9.94	20,78,294	13,954	43,121	
	14,839	1,585	3,26,825	9.20	246.95	5.95	390	247.05	247.90	325	25,500.00	1,235	28.50	28.65	585	-28.10	28.50	9.73	44,08,309	40,755	1,14,392	
	3,600	-1,196	1,85,026	9.16	207.30	3.10	130	207.45	208.20	715	25,550.00	2,015	38.80	38.95	325	-31.45	38.80	9.61	29,64,331	15,769	42,511	
	28,126	-1,514	11,75,386	9.14	171.00	-0.35	715	170.45	170.95	260	25,600.00	455	52.10	52.25	1,365	-35.50	51.95	9.49	60,26,817	25,382	95,679	
	21,780	-20,550	13,99,524	9.10	138.00	-4.15	390	137.50	138.00	845	25,650.00	845	68.85	69.10	585	-38.60	69.15	9.45	53,78,095	15,128	53,991	
	1,31,180	35,228	56,04,490	9.02	108.45	-7.90	390	108.25	108.45	455	25,700.00	260	89.65	90.00	1,105	-42.25	89.65	9.37	1,02,80,508	38,308	1,05,679	
	95,748	34,968	52,13,902	9.04	84.00	-10.05	260	84.00	84.30	2,015	25,750.00	390	114.60	115.00	9,295	-44.15	115.00	9.38	59,72,070	17,834	47,443	
	1,93,159	59,973	83,01,931	9.12	64.10	-11.45	585	64.05	64.35	780	25,800.00	390	144.55	145.10	650	-48.35	144.55	9.41	69,38,209	32,005	82,127	
	1,03,747	58,333	54,00,623	9.19	47.95	-12.00	455	47.95	48.20	845	25,850.00	520	178.20	178.80	780	-46.35	178.50	9.53	30,67,692	15,287	24,203	
	1,69,551	87,300	68,16,151	9.28	35.30	-11.85	195	35.05	35.30	1,430	25,900.00	260	215.05	215.95	260	-47.05	215.20	9.59	22,45,369	10,880	29,166	
	80,593	36,826	37,89,682	9.32	25.15	-12.15	1,690	25.10	25.30	975	25,950.00	260	255.55	256.45	325	-46.25	255.10	9.69	4,71,220	2,784	6,000	
	2,24,296	27,906	63,41,925	9.47	18.20	-11.65	845	18.05	18.25	780	26,000.00	65	298.00	298.90	325	-45.55	298.00	9.90	6,57,669	-1,847	32,160	
	66,387	23,792	27,44,285	9.57	12.75	-10.30	585	12.65	12.75	130	26,050.00	260	342.45	343.80	260	-42.30	343.50	10.27	88,501	-165	3,800	
	1,53,860	80,273	36,20,736	9.68	8.85	-9.25	2,990	8.85	9.00	6,00,795	26,100.00	260	388.25	389.50	130	-40.80	388.80	10.38	93,828	-1,643	9,229	
	76,300	37,753	20,07,951	9.97	6.60	-7.70	1,495	6.50	6.60	2,600	26,150.00	260	435.45	437.85	260	-35.45	437.65	11.11	15,140	-11	2,143	
	1,61,299	53,512	30,05,958	10.34	5.15	-6.15	780	5.00	5.15	1,950	26,200.00	260	483.75	486.05	260	-43.05	483.30	10.78	32,209	449	15,375	
	76,278	27,325	17,11,361	10.71	4.05	-5.00	65	4.00	4.05	975	26,250.00	130	532.30	535.75	130	-37.30	534.90	12.23	4,239	117	1,634	
	1,04,546	24,741	18,92,738	11.16	3.35	-4.15	325	3.40	3.45	195	26,300.00	260	581.60	584.35	195	-38.05	583.85	12.78	9,719	-114	6,579	
	64,656	25,928	11,42,694	11.64	2.85	-3.30	975	2.85	2.95	1,365	26,350.00	130	630.80	635.95	455	-34.90	632.75	13.23	1,178	54	903	

- **Example:**
- Suppose the CMP is 25694
- In the option chain, the CE OI is very high at the 26,000 strike.
- This suggests that 26,000 could act as a resistance area.
- If price moves toward this level, the chances of a pullback may be higher.
- Therefore, a CE sell trade could benefit if the market fails to break above this level.

- Similarly, high PE OI below the CMP can indicate a support zone.
- Since put OI is often concentrated in **strikes below the CMP**
- The highest PE OI is usually seen in **OTM or near-ATM put strikes**.

- **Change in Open Interest (Change in OI)**
- Change in Open Interest (OI) refers to the increase or decrease in the total number of outstanding futures or options contracts that are still open in the market.
- It shows whether new positions are being added or existing positions are being closed during a specific period.
- If Change in OI is positive → new positions are being created (more contracts opened).
- If Change in OI is negative → positions are being closed (contracts squared off or exited).

Option Greeks

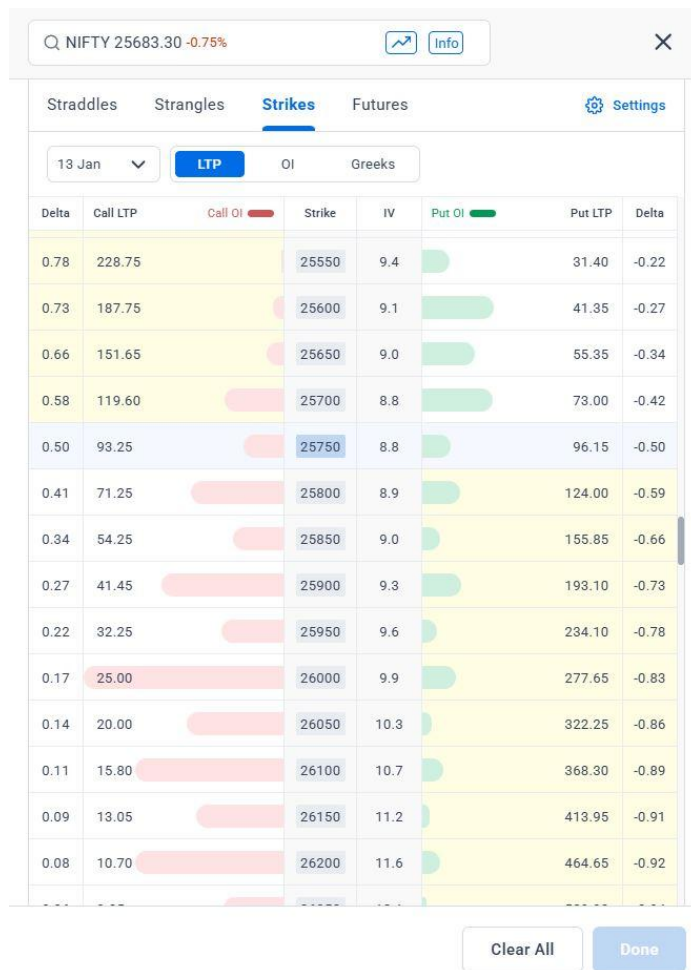
Option Greeks / Parameters

- Option Greeks are a set of mathematical measures used quantify different aspects of options pricing and behavior.
- They help traders and investors understand how changes in various factors affect the value and risk of options.
- The key option Greeks are: Delta, Gamma, Vega, and Theta, Rho.

1.) DELTA

- Delta measures the sensitivity of an option's price to changes in the underlying asset's price. It indicates the expected change in the option's value for a ₹1 change in the underlying asset.
- Delta ranges from 0 to 1. Whenever the value of underlying asset moves up or down, the premium value of the corresponding Option contract also changes as per the below calculation:

- More details can be checked in Sensibull Option Chain.



- **Example:** Delta Effect on Option Premium
- **Contract:** NIFTY 13 JAN 25750 CE @ 93.25
- **Current Market Price (CMP)** of NIFTY: 25780
Delta = 0.5

Now, suppose the CMP moves to 25880 from 25780 (i.e., it increases by 100 points).

- Change in Premium = Delta $\times$ Change in Underlying
 $= 0.5 \times 100 = 50$

So, New Premium = $93.25 + 50 = 143.25$

- Delta Value is always between 0 and 1. From ATM to OTM, Delta value decreases from 0.5 to 0
- From ATM to ITM, Delta value increases from 0.5 to 1
- Delta value is always equal to or near 0.5 at ATM
- Why you should buy ATM and not OTM?

- Three traders' scenarios are as given below:

CMP 25680 Increased by 100 point

Trader	Strike (CE)	Option Type	Premium (₹)	Delta	Profit (₹)	Total (₹)
Trader 1	25,550	ITM	229	0.8	80	309
Trader 2	25,750	ATM	93	0.5	50	143
Trader 3	26,000	OTM	25	0.2	20	45

- As per the illustration, Trader 1 has the highest profit, but also the highest risk if the price drops by 100. On the other hand, Trader 3 has lower profit but also lower risk in case of a price fall.
- Trader 2, who took the ATM option, has average profit and average risk.
- If the price goes up, profit increases; if the price goes down, the loss increases.

2.) GAMMA

- Shows **how much Delta will change** if the underlying price changes by ₹1.

$$\text{New Delta} = \text{Old Delta} + (\text{Gamma} \times \text{Price Change})$$

- More details can be checked in Sensibull Option Chain

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Call Delta	Strike	IV	Put Delta	Call and Put Greeks		
				Theta	Vega	Gamma
0.71	25550	10.9	-0.29	-7	18	0.0006
0.68	25600	10.8	-0.32	-7	19	0.0006
0.65	25650	10.7	-0.35	-7	20	0.0007
0.61	25700	10.6	-0.39	-7	20	0.0007
0.58	25750	10.5	-0.42	-7	21	0.0007
0.54	25800	10.5	-0.46	-7	21	0.0007
0.51	25850	10.4	-0.49	-7	21	0.0007
0.47	25900	10.4	-0.53	-7	21	0.0007
0.43	25950	10.4	-0.57	-7	21	0.0007
0.40	26000	10.3	-0.60	-7	20	0.0007
0.36	26050	10.4	-0.64	-7	20	0.0007
0.33	26100	10.4	-0.67	-7	19	0.0007
0.30	26150	10.4	-0.70	-6	18	0.0006
0.27	26200	10.4	-0.73	-6	17	0.0006

EXAMPLE

Strike: 25,800 CE Call Delta : 0.54

Gamma : 0.0007

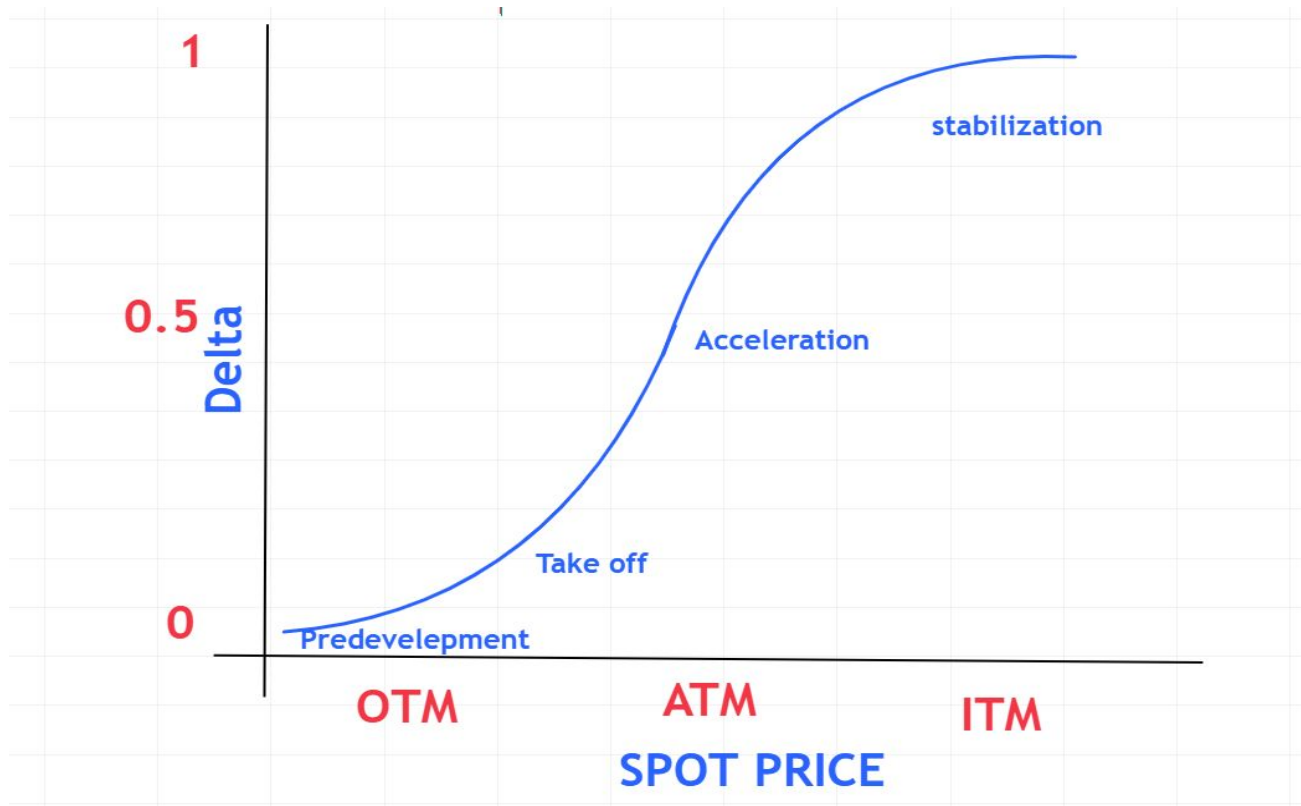
Price moves **+100 points**

New Delta = Old Delta + Change in Delta

$$\begin{aligned}\text{Change in Delta} &= \text{Gamma} \times \text{Price Change} \\ &= 0.0007 \times 100 = 0.07 \\ &= 0.0007 \times 100 = \mathbf{0.07}\end{aligned}$$

$$\mathbf{\text{New Delta} = 0.54 + 0.07 = 0.61}$$

DELTA VS SPOT



Strike Selection Logic

Buying rules

-  Avoid **deep OTM buying** → cheap but low movement
-  Buy **slightly OTM or ATM** → best risk–reward
-  Buy **ITM** if you want safer movement and can pay higher premium

Selling rules

-  Don't sell **ATM or ITM** options hoping they expire worthless
-  Sell **OTM** options if holding till expiry

One-line memory trick:

Buy near ATM for speed,
Buy ITM for safety,
Sell OTM for probability.

3.) VEGA

- **IV (Implied Volatility):** Implied Volatility is the market's forecast of the likely movement in a security's price. IV is used to price option contracts — higher IV leads to higher premiums, and lower IV leads to lower premiums.
- **Supply and demand** and time value are major factors that determine IV.
- Vega measures how much the premium of an option will change for a 1% change in IV.

- More details can be checked in Sensibull Option Chain

Q NIFTY 25738.25 +0.21%  [Info](#) ×

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0.47	25900	10.4	-0.53	-7	21	0.0007
0.43	25950	10.4	-0.57	-7	21	0.0007
0.40	26000	10.3	-0.60	-7	20	0.0007
0.36	26050	10.4	-0.64	-7	20	0.0007
0.33	26100	10.4	-0.67	-7	19	0.0007
0.30	26150	10.4	-0.70	-6	18	0.0006
0.27	26200	10.4	-0.73	-6	17	0.0006

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- **Vega Calculation Example – NIFTY 25800 CE**

- $CMP = 25,738$
- $\text{Option Premium} = 50$
- $IV = 10.5\%$
- $Vega = 21$

$IV: 10.5\% \rightarrow 11.5\% (+1\%)$

- If IV increases by 1%, the option premium will increase by the Vega value:
- $\text{New Premium} = 50 + 21 = 71$

Important Trading Insights

- Buying options when **IV is low** is better (Vega works in your favor)
- Selling options when **IV is high** is better
- Near expiry → **Vega reduces**
- Big events (news, RBI, budget) → IV rises → Vega impact increases

Vega measures how much option price changes due to volatility, not price movement.

d) THETA

- Theta is also called Time Decay, and it is a very important parameter. The value of an option contract premium decreases as time passes and eventually becomes zero when the contract expires.
- Theta depends on each passing day and is purely based on time. If the number of days to expiry is less, the Theta value is high. The rate of decay will be higher as the contract nears expiry.

- In a consolidating or ranging market, the premium will reduce by the Theta value.
- If a trader chooses to buy an option contract and the market remains in consolidation, the trader stands to lose capital to the extent of **Time Decay (Theta)**.
- Therefore, option sellers benefit more than option buyers in a ranging market.

- More details can be checked in Sensibull Option Chain

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- **Theta calculation Example – NIFTY 25800 CE**
- **CMP = 25738**
- **Premium = 50**
- **Theta Value = -7**

Day 1 → Day 2 (No price movement, no IV change)

Today's premium: ₹ 50

Tomorrow's premium = $50 - 7 = 43$

Option loses ₹7 in one day only due to time decay

- The probability of making a profit for a call option buyer is considered as 33.3%, as the trader can lose in two situations:

When the market is in consolidation (time decay loss)

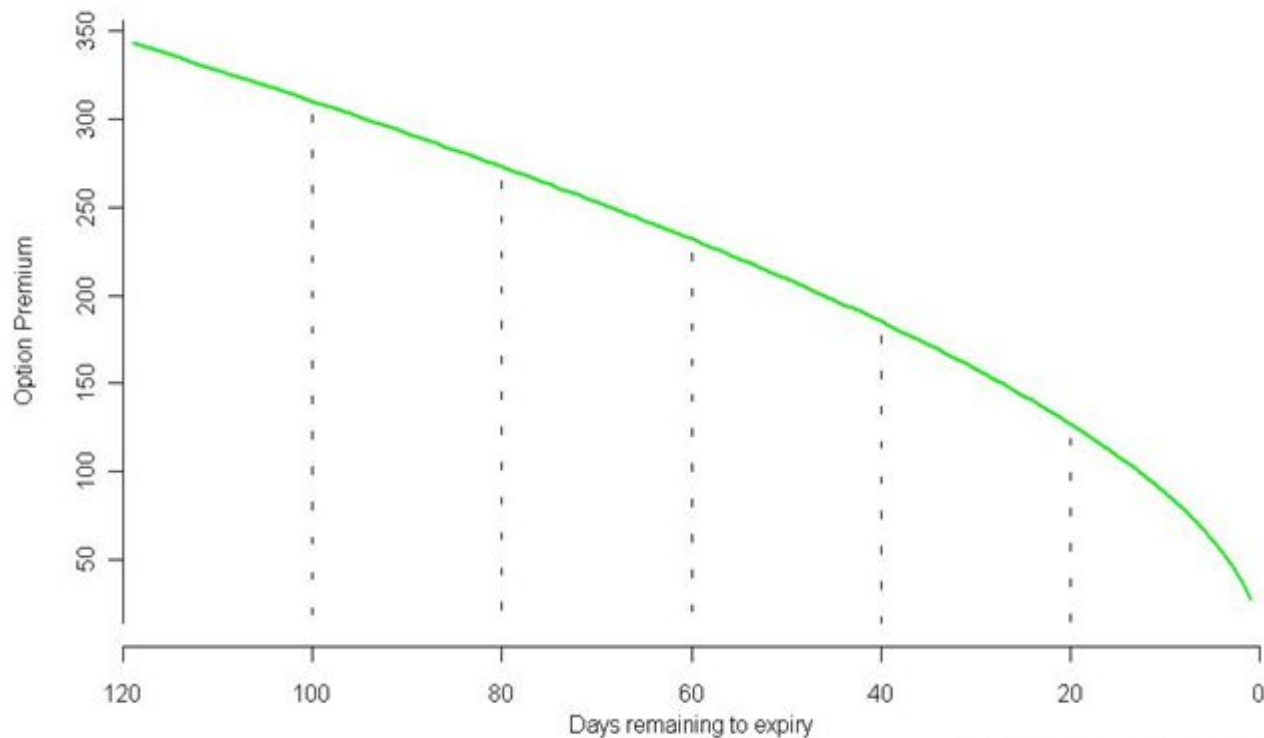
When the market goes down

- The option call option seller is considered to have around 66.6% probability of making a profit, as they can gain from both:

Price fall

Consolidation (due to Theta / Time Decay)

Option premium vs time to expiry



e) RHO

- Shows **how much option premium changes if interest rate changes** by 1%.
- More important for **longer-term options** than intraday/weekly.

THANK YOU